Week 10: Multilevel regression and post-stratification

23/03/26

1 Overview

In this lab we will look at an example of doing multilevel regression with the fertility intentions data. We will also go through how to do post-stratification and MRP.

2 Packages and data

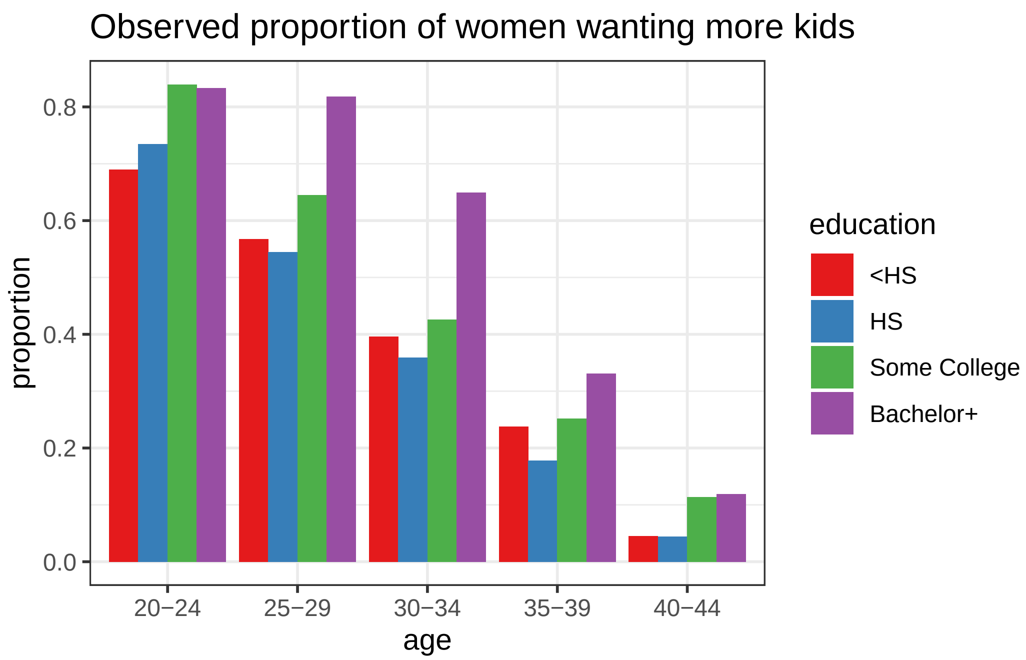
Read in the data from NSFG:

```
library(tidyverse)
library(lme4)
library(rstan)
d <- read_csv("../data/intentions.csv")
d <- d |>
  mutate(educ_cat = ifelse(edu == "Bachelor"|edu == "Post-Grad", "Bachelor+", edu)) |>
  mutate(educ_cat = fct_relevel(educ_cat, "<HS", "HS", "Some College", "Bachelor+"))
```

3 Descriptives

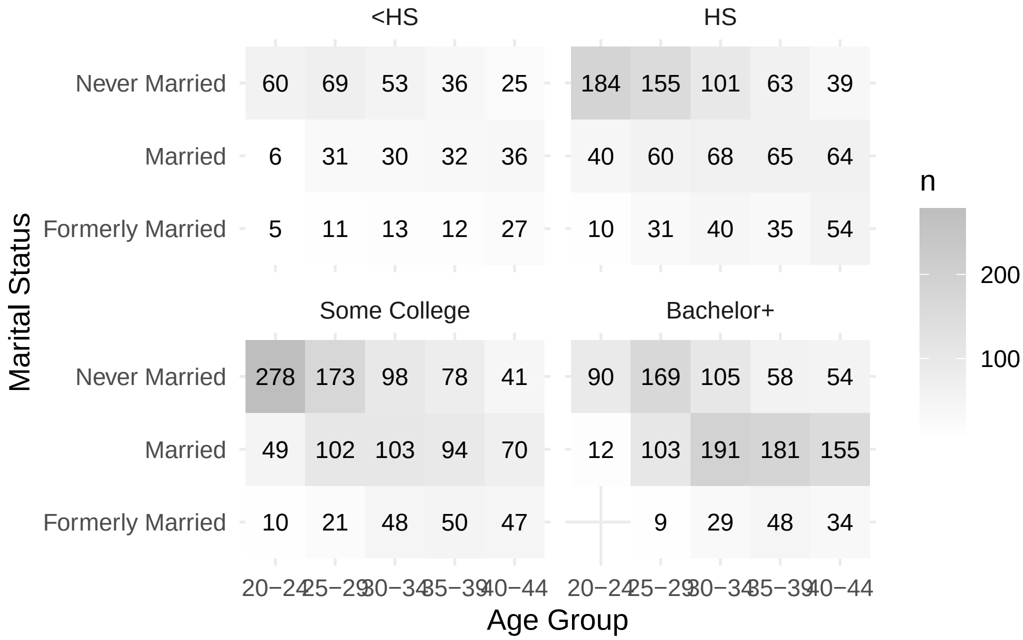
Let's do some brief descriptive plots. Here's the observed proportion of women wanting more kids by edication and age:

```
d |>
  group_by(age_gp, educ_cat, wants_more_kids) |>
  tally() |>
  group_by(age_gp, educ_cat) |>
  summarize(prop_wants_more = n[wants_more_kids==1]/sum(n), n = sum(n)) |>
  filter(age_gp!="15-19") |>
  ggplot(aes(age_gp, prop_wants_more, fill = educ_cat)) +
  labs(title = "Observed proportion of women wanting more kids", x = "age", y = "proportion")
  theme_bw()+
  scale_fill_brewer(palette = "Set1", name = "education")+
  geom_bar(stat = "identity", position = "dodge")
```



Here's a tile plot of the sample sizes:

```
d |> group_by(age_gp, marital_formal, educ_cat) |>
  summarize(n = n()) |>
  filter(age_gp!="15-19") |>
  ggplot(aes(x=age_gp, y=marital_formal, fill=n)) +
  geom_tile() +
  scale_fill_gradient(low="white", high="grey") +
  geom_text(aes(label=n), size=3) +
  xlab("Age Group") + ylab("Marital Status") + theme_minimal()+facet_wrap(~educ_cat)
```



3.1 Question

Do another interesting descriptive chart of the data.

4 Multilevel model

Let's fit the same multilevel model as seen in the lecture using the `lme4` package

```
resp4 <- d |> filter(age_gp!="15-19")  |> drop_na() |> mutate(age_gp = as_factor(age_gp)) |>
  mutate(age_gp = fct_relevel(age_gp, c("20-24", "25-29", "30-34", "35-39", "40-44")))

mod <- glmer(wants_more_kids ~ marital_formal + (1|age_gp) + (1|educ_cat), data = resp4, fam
summary(mod)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: wants_more_kids ~ marital_formal + (1 | age_gp) + (1 | educ_cat)
   Data: resp4

      AIC       BIC    logLik -2*log(L)  df.resid
```

```
   4331.2    4362.6   -2160.6    4321.2      3950

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.9035 -0.7395 -0.2314  0.6866  5.0382

Random effects:
 Groups    Name        Variance Std.Dev.
 age_gp    (Intercept) 1.7471   1.3218
 educ_cat  (Intercept) 0.2297   0.4793
Number of obs: 3955, groups:  age_gp, 5; educ_cat, 4

Fixed effects:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                  -0.7963     0.6483  -1.228   0.2193
marital_formalMarried         0.3071     0.1261   2.436   0.0148 *
marital_formalNever Married   0.6298     0.1247   5.050 4.41e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) mrtl_M
mrtl_frmlMr -0.148
mrtl_frmlNM -0.153  0.771
```

Note! This can also be done in Stan, in a Bayesian framework. This is useful if you want to e.g. play around with the structure of the hierarchical model, which is much easier to control in this framework.

```r
stan_data1 <- list(N = nrow(resp4),
                  A = nlevels(resp4$age_gp),
                  E = nlevels(resp4$educ_cat),
                  age = as.numeric(resp4$age_gp),
                  edu = as.numeric(resp4$educ_cat),
                  x1 = as.numeric(resp4$marital_formal == 'Married'),
                  x2 = as.numeric(resp4$marital_formal == 'Never Married'),
                  y = resp4$wants_more_kids)

mod1 <- stan(data = stan_data1,
             file = "../code/models/intentions.stan",
             iter = 1000,
             seed = 243,
```

```r
                refresh=0)
summary(mod1)$summary[c("beta0", "beta1", "beta2", "sigma_a", "sigma_e"),
                        c("mean", "se_mean", "n_eff", "Rhat")]
```

```
              mean     se_mean     n_eff     Rhat
beta0   -0.5198190 0.026874666  456.5111 1.002960
beta1    0.2894595 0.003329329 1311.2561 1.000565
beta2    0.6121447 0.003338259 1271.7130 1.002065
sigma_a  1.4035783 0.010265429 1411.0734 1.002603
sigma_e  0.6696694 0.011445434  696.2899 1.001549
```

To pull out the estimates from `lme4`:

```r
coef(mod)
```

```
$age_gp
      (Intercept) marital_formalMarried marital_formalNever Married
20-24   0.8455578             0.3070844                   0.6297794
25-29   0.1509033             0.3070844                   0.6297794
30-34  -0.5704127             0.3070844                   0.6297794
35-39  -1.5671327             0.3070844                   0.6297794
40-44  -2.8334241             0.3070844                   0.6297794

$educ_cat
             (Intercept) marital_formalMarried marital_formalNever Married
<HS           -1.1165369             0.3070844                   0.6297794
HS            -1.1969670             0.3070844                   0.6297794
Some College  -0.7309838             0.3070844                   0.6297794
Bachelor+     -0.1398147             0.3070844                   0.6297794

attr(,"class")
[1] "coef.mer"
```

```r
fixef(mod)
```

```
                (Intercept)       marital_formalMarried
                 -0.7963065                   0.3070844
marital_formalNever Married
                  0.6297794
```

```
ranef(mod)
```

```
$age_gp
      (Intercept)
20-24   1.6418643
25-29   0.9472098
30-34   0.2258938
35-39  -0.7708262
40-44  -2.0371176

$educ_cat
             (Intercept)
<HS          -0.32023040
HS           -0.40066049
Some College  0.06532273
Bachelor+     0.65649180

with conditional variances for "age_gp" "educ_cat"
```

4.1 Question

Compare the multilevel model with just a standard logistic regression. Which model has a higher AIC?

5 Post stratification

We are interested in obtaining estimates for the proportion of women who want more kids by age. ## Raw proportions by age

Before we do any post-stratification, we can just calculate the raw survey proportions by age:

```
raw_age <- resp4 |>
  mutate(edu = ifelse(edu == "Bachelor"|edu=="Post-Grad", "Bachelor+", as.character(edu))) |>
  group_by(age_gp) |>
  summarize(proportion = mean(wants_more_kids, na.rm = TRUE)) |>
  mutate(estimate = "raw")
```

5.1 Post-stratification dataset

We are going to use data from the American Community Survey (ACS) as the post-stratification dataset. Note that I got these data from IPUMS USA. Load in ACS data:

```
cell_counts <- read_csv("../data/acs_counts.csv")
```

5.2 Question

Compare proportions by age/education/marital group in ACS versus survey.

5.3 Standard post-stratification

Calculate the proportion by group (age, education, marital status) and then re-weight based on the ACS data:

```
ps_age <- resp4 |>
  mutate(edu = ifelse(edu == "Bachelor"|edu=="Post-Grad", "Bachelor+", as.character(edu))) |
  group_by(age_gp, edu, marital) |>
  summarize(prop = mean(wants_more_kids, na.rm = TRUE)) |>
  filter(marital !="Cohabitating") |>
  left_join(cell_counts) |>
  mutate(prod = prop*n) |>
  group_by(age_gp) |>
  summarize(proportion = sum(prod)/sum(n)) |>
  mutate(estimate = "poststrat")
```

5.4 MRP

Get predicted proportions for each group:

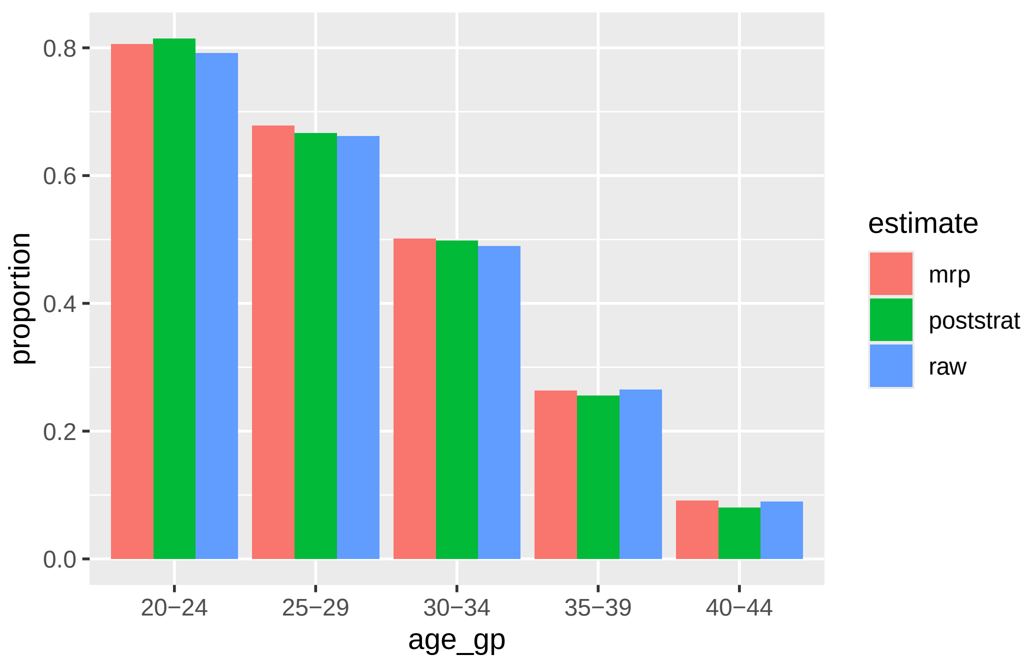
```
age_groups <- rownames(coef(mod)$age_gp)
educ_groups <- rownames(coef(mod)$educ_cat)
marital_groups <- c("Formerly Married","Married", "Never Married")

pred_data <- expand_grid(age_gp = age_groups, educ_cat = educ_groups, marital_formal = marit

pred_mrp <- plogis(predict(mod, pred_data))

pred_prop <- cbind(pred_data, pred_mrp)
```

Now post-stratify by age:

```
mrp_age <- pred_prop |>
  rename(edu = educ_cat, marital = marital_formal) |>
  left_join(cell_counts) |>
  mutate(prod = pred_mrp*n) |>
  group_by(age_gp) |>
  summarize(proportion = sum(prod)/sum(n)) |>
  mutate(estimate = "mrp")
```

Compare on a plot:

```
ps_age |>
  bind_rows(mrp_age) |>
  bind_rows(raw_age) |>
  ggplot(aes(age_gp, proportion, fill = estimate))+
  geom_bar(stat = "identity", position = "dodge")
```



5.5 Question

Do the same thing but compare across education groups.